

E-Payment System to Reduce Use of Paper Money for Daily Transactions

Abstract—This paper presents a framework of an E-payment system which can be used for daily petty cash transactions. The framework involves communication of encrypted messages through an insecure network using Advanced Encryption System (AES) with a timestamp key. To ensure the message authentication and integrity, the hash-based message authentication code (HMAC) is used. The associated key in HMAC is considered as a function of the phone MAC address as well as the user's biometric data for added security. Timestamp key is updated each time a transaction takes place. The framework can be implemented to replace the use of paper money for daily transactions.

Index Terms—E-payment system, Security, Cryptography

I. INTRODUCTION

E-payment system is one of the elements to simplify life in the abundance of internet facilities of modern era. The system has the potential to reduce the use of paper money which gets old and damaged due to human contacts and age. Furthermore, paper money costs for its printing and is vulnerable to theft and robbery. Often, old, slightly torn or burnt money is not accepted for use.

Thus, E-payment system is a great alternative, resolving this disadvantages and demerits associated with paper money [1-2]. Different forms of E-payment systems such as internet banking services are widely used in Bangladesh. Moreover, phone-banking systems such as bKash, Nagad and Rocket are also available as E-payment platforms [3-4]. Fig. 1 illustrates the serviceability of an e-payment system with one which can transfer money to another account/card holder in the system. A person can have his own bank account together with an E-payment system/card. Thus, consumers can use the system in a flexible way for paying shopping, transportation, electricity, water bills, etc.

Security of the e-payment card or system should be strong. If the card/device is lost, the hackers should be unable to use, retrieve or transfer the money from the card/device [5-7]. In case of the loss of card/device, the user should be able to stop any transaction or ground the activities as soon as possible. Additionally, such payment systems are required to maintain

security standards for mobile devices including smartphones as discussed further in [8-9].

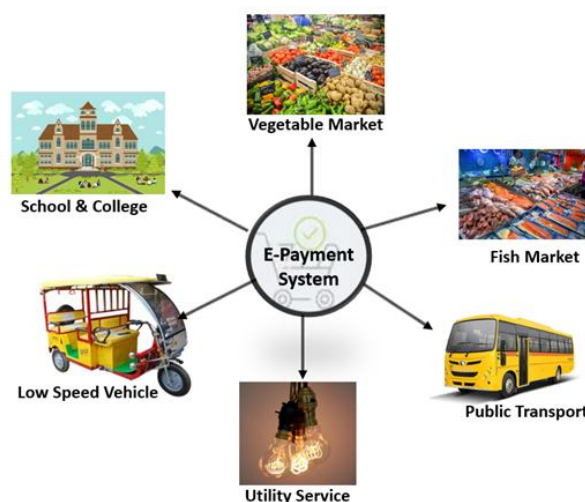


Fig. 1. Illustration of e-payment system applications.

A survey paper by Masihuddin et al. explains a model architecture for payment gateway which involves five participants: customer, client's bank, merchant, merchant's bank, and the payment server. The customer shops online and uses the online e-payment system to pay for the merchandise. Through the gateway, the connection is established with the bank or credit card organization to verify the account details through the bank or the credit card organization. The authors of this survey did not provide any details on the security aspects of this model. In addition, the paper provided a comparison between 11 different payments in terms of their service features such as fees, worldwide availability, card types, bundle accounts, and costs; however, no details on security aspects were presented [12].

The work of Bogdan-Alexandru [13] discusses security issues and solutions in e-payments systems. The author presented the types of e-payment systems and the security issues that threaten the e-payment systems, i.e. the different types of attacks that an e-payment system could be prone to.

The paper lacks the discussion of e-payment architecture/frameworks and details on security components. The author recommended that e-payment systems should have reliable and secure methods with appropriate authentication to protect the system from any risks that could jeopardize the clients, merchants, banks and credit cards data.

In [14] Yadu and Sharma, presents a study to identify the issues and challenges of electronic payment systems with solutions to improve the quality of e-payment systems. The study lacks the discussion on the inner details of e-payment security architecture/framework. From the research in the literature there are not enough details available on e-payment systems in relation to the security frameworks or methods/techniques adapted in such systems. In this work, we propose a detailed framework on the security aspects of e-payment systems.

In the proposed framework, the user's national identity card (NID) or birth registration certificate (BRC) is used for generating the user registration number with the bank authority for authentication and initiating the user's account on the e-payment system. Advanced Encryption System (AES) [10] with a timestamp key is employed. To ensure message authentication and integrity, the hash-based message authentication code (HMAC) [11] is used. The associated key in HMAC is considered as a function of the phone MAC address and biometric user fingerprint (B_p). The timestamp key is updated each time a transaction takes place. An attacker will not be able to read the message as the message is encrypted with a secret key. Any slight modification in the message content will change the whole message. Therefore, any attacker will not be able to change the actual message content. Another problem includes the replay attack, in which an attacker sends the message to the server again later at his/her convenient time is prevented as a timestamp key is used and the key is modified each time a transaction is successful.

The rest of the paper is organized as follows. In section II, we present the framework of the proposed E-Payment system. Section III presents analysis and discussion of the framework. The conclusion of the present work is presented in section III.

II. E-PAYMENT SYSTEM

Figs. 2-3 illustrate the framework of the proposed E-Payment system. Before proceeding with the transaction, first, registration for the E-payment system for a user is done through a bank officer, upon verifying the user's NID or BRC. Specifically, it works as follows. The E-payment system is installed on the user's mobile device. Upon successful completion of the installation, the system will require an activation code from the bank officer. The activation code and the NID or BRC are used to generate the private key K_1 which is used in HMAC applied on the user's message M as shown in Figs. 2-3. The officer provides a username, records user fingerprint (B_p) and facilitates to create the password K_2 which is only known by the user but not the officer.

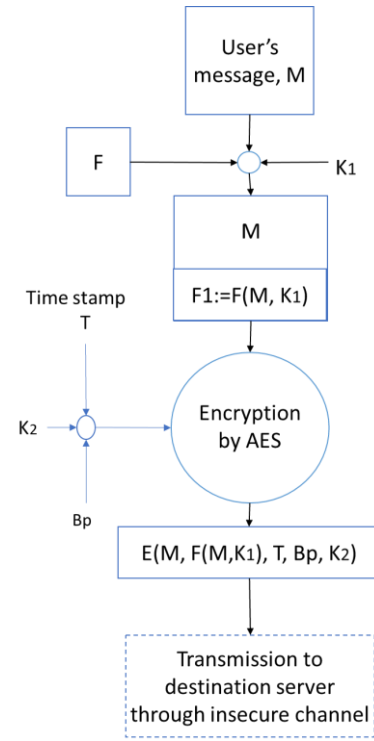


Fig. 2. Illustration of message encryption from user's side.

When a money transfer is required by the user, the user must log on the system with his/her username and password. Next, the user is required to enter the necessary details including, the receiver's username (given by the bank), and the amount of money in the system interface. Upon proceeding with the transfer, the user must input the fingerprint (B_p); next, the following tasks are performed:

The message M is passed through HMAC with the generated key K_1 to produce a hashed value $F1$. The message M and $F1$ are then combined and encrypted by user's password K_2 , user's fingerprint (B_p) and the timestamp T . The resulting Encrypted data is transmitted through the insecure channel to the bank server. The server receives the message and decrypts the message as illustrated in Fig. 3. The server uses the same key K_2 , the fingerprint (B_p) and the timestamp T to decrypt the message. Next, message integrity and the authenticity of the user's device are checked by hashing the message part M using HMAC with the key K_1 . Only if the hashed value $F2$ is the same as $F1$, then the message M is accepted. Once M is accepted, the amount of money S and the receiver's name are retrieved from M , and the name is verified in the database. Next, if the sender's balance is more than S , the sender's balance is updated by deducting S , while S is added to the receiver's account. Otherwise, the transfer is made futile. The message is sent to sender and the receiver for successful transfer. In case of an aborted transfer, only sender is informed of the reason by the server. Upon each successful transfer, the time stamp T is updated.

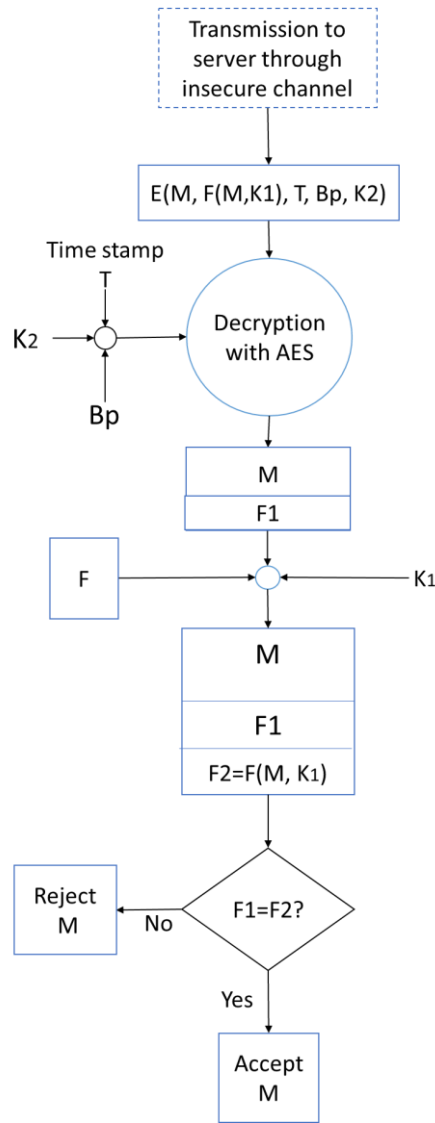


Fig. 3. Illustration of message decryption from server's side.

III. ANALYSIS AND DISCUSSION

AES is a robust cryptographic algorithm and any alteration of a single bit in the encrypted message will produce a completely different message after decryption. Furthermore, the key K_1 produced during application activation is used in HMAC which can be used by the bank to identify the user's device. At the same time, message integrity is confirmed. Another important aspect is the replay attack in which the attacker will resend the message to the server at his/her convenient time so that the server again transfers the money to the receiver. To prevent the replay attacks, the timestamp T is used and updated when a transaction is successful. A replay attack is averted using the proposed framework, since T is integrated used with K_2 and fingerprint (BP); and the resulting key is used with the AES process.

Essentially, the framework proposed in this paper has resulted in a new contribution to the domain of e-payment systems with main features and strengths that include:

- Support for small monetary transactions with a daily limit set upon user registration. This daily limit can be set as desired within a regulatory range specified at the time of registration.
- A simple model with reduced computational complexity by employing symmetric encryption in contrast to public-key cryptography.
- The use of mature and well-known hash functions (HMAC) and symmetric encryption algorithms (AES) that can be executed on mobile devices with low computational power and storage space.
- An enhanced approach that incorporates the users MAC address and biometric data that prevents monetary theft even when the thief may have obtained the user device and password credentials.
- The hashed MAC address using a one-way non-collision algorithm (HMAC) ensures that the transaction message is unique to the user device and could not have been sent from any other device. Hence, this mechanism eliminates the need for the one-time password (OTP) sent by the bank as found in many other e-payment systems.
- The framework prevents impersonation attacks as from the previous point in addition to replay attacks using timestamps for sequencing purposes.

A drawback of the approach was noted since the registration process had required the user to initially activate their account details at the authorized bank, following which the symmetric key value (K_2) would be provided to the user as required in all monetary transactions.

IV. CONCLUSION

A framework is proposed in this paper to reduce the use of cash for daily monetary transactions. Since cash/paper money grows old over time and may be damaged due to repeated handling, an E-payment system can offer a great alternative, providing the flexibility of payments for small transactions without requiring change of money. With such an E-payment system, users can pay transport fares, electric bills, tickets, and shopping bills.

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